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# 4 Star

## 1. What are the problems faced by Four Star Industries in their mattress manufacturing operations?

**Drivers (external)**

- Intensifying competition: Dreammakers, Dunlopillo, Simmons, Slumberland and late entrants
- Product proliferation required by dealers (e.g. Novena, 25% of sales)
- Dealers stopped holding stock and taking care of logitics: Four Star now has to carry the inventory, take care of delivery and is expected to deliver within 2 days
- Demand is seasonal and highly dependent on promotions (very uncertain during discounts)

**Operational**

- Models went from 13 to 230 (18×)
- Volume fell from 18,700 to 15,080 units
- Fill rate fell from 70% to 60%
- Warehouse space increased from 5,000 to 45,000 sq ft (9×)
- Stockouts and excess stock at the same time
- Scheduling problem:
  - Setup-heavy machines (coil spring machine and quilt panel stitching machine)
  - Fixed batches (100 spring units, 50 quilt panels, assembly batches of 9–12 or 3)
- Raw material MOQs (24 t of wire, 2,000–3,000 m of fabric per variety, 45 varieties) engender large cycle stocks
- Safety-stock rule that ignores variability (2 weeks of demand for every model)
- Orders taken by fax / phone
- No forecasts
- No commitments from dealers
- Expedited backorders

**Consequences**

- Operating losses for three years (revenue down from S$9,000,000 in 1989 to S$6,000,000 in 2002)
- Overtime, wastage, and long-serving staff with stress-related health problems threatening to quit

## 2. How does the proliferation of mattress varieties impact Four Star's operations?

**1. Demand is split**

Average demand per model decreased while total volume stayed the same

| Year | Models | Units | Units / model / week |
|---|---|---|---|
| 1996 | 13 | 18,700 | 27.7 |
| 2002 | 230 | 15,080 | 1.3 |

2002 volume:
- 20 fast movers sell 9,300 units
- 210 slow movers 6,300
- Total 15,600

The last of the 210 models represents 40% of the volume but 91% of the SKUs

**2. Variability per SKU explodes**

$$\mathrm{CV} = \frac{\sigma}{\mu}$$

Average CV:
- Fast movers:

$$\mathrm{CV}_{\text{fast}} = \frac{1}{20}\sum_{i=1}^{20}\frac{\sigma_i}{\mu_i} = 0.1$$

- Slow movers: $\mathrm{CV}_{\text{slow}} = 25/30 = 0.83$

A slow mover sells $30/52 = 0.58$ units per week, so a single assembly batch of 3 covers 5.2 weeks

**3. The safety-stock rule fails where products proliferated**

- Holding 2 weeks of average demand gives us $\mathrm{SS} = 2\mu/52$
- Over the 2-week protection interval the implied service level is $\mathrm{CSL} = \Phi\left(\mathrm{SS}\ /\left(\sigma\sqrt{T}\right)\right)$:

| Model | $\mu$ | $\sigma$ | CV | SS (2 wks) | $\sigma_T$ | Implied CSL |
|---|---|---|---|---|---|---|
| PU | 1200 | 108.1 | 0.09 | 46.2 | 21.2 | 98.5% |
| CMD | 920 | 114.5 | 0.12 | 35.4 | 22.5 | 94.2% |
| GOMK | 800 | 102.9 | 0.13 | 30.8 | 20.2 | 93.6% |
| JC | 650 | 89.9 | 0.14 | 25 | 17.6 | 92.2% |
| PL | 630 | 61.7 | 0.1 | 24.2 | 12.1 | 97.7% |
| SM2 | 580 | 53.1 | 0.09 | 22.3 | 10.4 | 98.4% |
| SJL | 540 | 43.6 | 0.08 | 20.8 | 8.6 | 99.2% |
| PU(S) | 500 | 38.8 | 0.08 | 19.2 | 7.6 | 99.4% |
| PF | 480 | 43.6 | 0.09 | 18.5 | 8.6 | 98.5% |
| RC | 440 | 34.2 | 0.08 | 16.9 | 6.7 | 99.4% |
| CMD(K) | 380 | 40 | 0.11 | 14.6 | 7.8 | 96.9% |
| OGN | 330 | 30.8 | 0.09 | 12.7 | 6 | 98.2% |
| PU(K) | 310 | 38.3 | 0.12 | 11.9 | 7.5 | 94.4% |
| SBS | 290 | 24.3 | 0.08 | 11.2 | 4.8 | 99% |
| GMK(K) | 250 | 22.5 | 0.09 | 9.6 | 4.4 | 98.5% |
| SC788 | 230 | 18.5 | 0.08 | 8.8 | 3.6 | 99.3% |
| GA | 210 | 19.5 | 0.09 | 8.1 | 3.8 | 98.3% |
| USQ | 200 | 19.6 | 0.1 | 7.7 | 3.8 | 97.7% |
| SC1116 | 190 | 17.3 | 0.09 | 7.3 | 3.4 | 98.4% |
| EX | 170 | 21.8 | 0.13 | 6.5 | 4.3 | 93.7% |
| **Each slow mover** | 30 | 25 | 0.83 | 1.15 | 4.9 | **59.3%** |

- Fast movers are over-protected (CSL > 93%)
- Slow movers are a coin flip

**4. Order fill rate collapses on multi-line orders**

- A dealer order is filled only if every line is in stock
- If we have a 59% availability per slow-moving line:

| Slow-moving lines in order | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| P(order complete) | 59% | 35% | 21% | 12% | 7% |

An order containing a few slow movers is almost certain to wait (bottleneck)

**5. Raw-material and setup effects**

Every new model needs its own spring-unit or quilt setup and sometimes its own fabric or wire gauge. MOQs are per material, not per model (a fabric MOQ of 2,000 m is 250 panels, i.e. 8.3 years of demand for a slow mover. This increases warehouse space and locks cash up in stock.

## 3. Analyze the various courses of action that Sia Meng is contemplating to solve Four Star's operational problems. How effective will these actions be?

Sia Meng is considering 4 ideas:

**a) Reduce the number of models back to the levels in 1996 or 1998.**
Pooling slow-moving in $N$ models, will reduce total safety stock (SS) by a factor of $\sqrt{N}$ (as per the square-root law) - assuming customers switch models.

This means, going from the 230 models to 80 models (1998) will reduce it by 41%. Going to 13 models will reduce it by 76%. The problem is, the dealers are demanding for their own exclusive models and they are most likely to walk away if they don't get what they want.

Novena sales alone constitute 25% of sales (≈ S$1,500,000) and losing them may cost more than the inventory saved.

Also, the 13 models are fewer than today's 20 fast moving mattresses and going back to 1996 numbers would hurt best selling mattresses.

Therefore, this solution is only **Partially effective**. The company should remove only the unprofitable models, and should avoid making blanket cuts.

**b) Make dealers order minimum number corresponding to batch size.**
Slow moving models have a batch of 3 which is 5.2 weeks of demand. Implementing this proposal forces the stock onto dealers, who decided not to carry inventory anymore. So, the solution risks the dealers moving to competition.

Also, even if the dealers were to accept the proposal, this would not reduce inventory in the overall supply chain. It only moves it to dealers, and that would not be an effective solution.

This proposal also does not solve the batching issue in the production process. The spring units batch (100) and quilt panels batch (50) depend on the equipment. Imposing a minimum order size would not address this issue. Therefore, the solution will **Not be effective**.

**c) Reduce safety stock requirement for finished mattresses.**

$$\mathrm{SS}_{\mathrm{week}} = \frac{2 \times 9300}{52} + \frac{2 \times 6300}{52} = 600$$

$$\mathrm{SS}_{\mathrm{week}} \times 150 = 90000$$

The current inventory policy translates to 600 units of Safety Stock worth (S$90,000), with an annual holding cost of S$18,000 at 20%. These numbers are relatively small and would not resolve the core issue.

Slow moving products are normally in stock 59% of the time. A reduction of Safety Stock from 2 to 1 week drops slow-movers' CSL from 59% to 55%. Since, fill rate already was down to 60% from 70%, this solution will make it worse.

The core issue is not safety stock but the fact that the warehouse is filled with cycle stock, fabric bought at MOQ, and defective finished goods.

So, this solution will **Not be effective**. But, calculating SS using $\sigma$ instead of weeks of demand will improve the numbers.

**d) Relocating manufacturing Overseas.**
While offshoring may reduce overtime and facility costs, does not address product complexity, scheduling issues and inventory mismatch. Besides, Four star's Labour and capacity are not the bottlenecks.

Taking manufacturing overseas will increase lead time, which may compromise their ability to meet the dealer's expectation of "2-day delivery. It will also need a higher inventory level (from 2 to 5 weeks) and safety stock will go up by $\sqrt{5/2} = 1.58\times$.

Also, mattresses are bulky andcostly to ship, and this will increase the costs for the products in an already very competitive market.

Therefore, this solution is **Not effective**.

## 4. Discuss possible ways in which Four Star can reconfigure its product design, manufacturing and order fulfilment. How will this help to solve its operational problems?

By maintaining the diversity for the dealer without holding inventory in the warehouse, we can: standardize parts, differentiate late, and only hold stocks where demand is consolidated.

### 4.1 Product design: modular platform

The differentiation would come from the spring unit (count × gauge × dimensions), quilt fabric, and foam. The 20 fast-moving items already utilize only a limited number of spring units:

| Spring unit (size-count-gauge) | Fast models | Fast demand | Slow models (≈ $kn$) | Total demand |
|---|---|---|---|---|
| Q-788-2.1 | 7 | 4,210 | 73.5 | 6,415 |
| Q-1116-2.1 | 4 | 2,470 | 42 | 3,730 |
| S-788-2.1 | 1 | 500 | 10.5 | 815 |
| K-1116-2.1 | 1 | 380 | 10.5 | 695 |
| Q-688-2.1 | 2 | 620 | 21 | 1,250 |
| K-788-2.1 | 2 | 560 | 21 | 1,190 |
| Q-1308-1.7 | 1 | 200 | 10.5 | 515 |
| Q-1116-1.9 | 2 | 360 | 21 | 990 |

230 models are built from about 8 spring units.

Recommendations:

- **Standardize the spring unit platform**: the company will manufacture spring units with only one gauge (2.1 mm) and 3–4 spring counts per size. The 1.9 mm and 1.7 mm gauges are only used to manufacture 1,505 mattresses per year, while the 2.1 mm gauge is used to manufacture 14,095 mattresses per year. Both have the same minimum order quantity of 24 t. Therefore, an order of 1.9 mm wire lasts approximately 14 times longer than an order of 2.1 mm wire.

- Rationalize fabrics (45 → ~12–15). With MOQ = 2,000–3,000 m and 8 m per panel, fabric cover per order is:

| Fabric | Panels / yr | m / yr | MOQ (m) | Weeks of cover | Avg cycle stock (S$) |
|---|---|---|---|---|---|
| QF1 | 5,180 | 41,440 | 3,000 | 3.8 | 4,699 |
| QF2 | 1,930 | 15,440 | 3,000 | 10.1 | 6,310 |
| QF3 | 965 | 7,720 | 2,000 | 13.5 | 3,133 |
| QF4 | 1,840 | 14,720 | 2,000 | 7.1 | 4,207 |
| QF5 | 855 | 6,840 | 2,000 | 15.2 | 2,417 |
| QF6 | 795 | 6,360 | 2,000 | 16.4 | 3,580 |
| QF7 | 755 | 6,040 | 2,000 | 17.2 | 5,370 |
| QF8 | 1,250 | 10,000 | 2,000 | 10.4 | 4,833 |
| QF9 | 525 | 4,200 | 2,000 | 24.8 | 4,296 |
| QF10 | 515 | 4,120 | 2,000 | 25.2 | 6,265 |
| QF11 | 505 | 4,040 | 2,000 | 25.7 | 4,207 |
| QF12 | 485 | 3,880 | 2,000 | 26.8 | 7,518 |

The 12 fabrics in the fast-mover BOMs can cover all models according to the case's own BOM assumption. The other 33 varieties each tie up an average of 1,000 m of fabric, which is equivalent to approximately S$146,198 of fabric and S$29,240 per year to hold, for low-volume models where a single MOQ can cover several years of demand.

- Give dealers exclusivity through low-cost, late features, such as label, tape edge colour, quilting pattern, foam/comfort layer, name and price point. All models would be built on standard spring units and a reduced fabric palette. Novena keeps its "Premier" range, but for the factory it is simply five variants of one platform.

**4.2 Manufacturing: hybrid make-to-stock / assemble-to-order**

Move the **decoupling point**:

- Fast movers (A items, 60% of volume): make-to-stock (MTS) of finished mattresses. Safety stock for fast movers is set with the formula $\mathrm{SS} = z\sigma\sqrt{T}$, rather than a fixed two weeks of demand.
- Slow movers: assemble-to-order (ATO). Safety stock for slow movers is held as spring units and quilt panels rather than finished mattresses, pooled across all the models that share them. Foam is bought to order, with a 2-day lead time. Assembly is manual with no significant setup, and the three assembly tables have spare capacity, since current volume is 19% below 1996. Slow-mover volume is also low, at only 24 mattresses per day, so they can be assembled within the 2-day delivery window.

Component pooling: if $n$ independent models share the same component, the pooled standard deviation is $\sigma_{\text{pool}} = \sigma\sqrt{n}$ instead of $n\sigma$, so total safety stock falls by a factor of $\sqrt{n}$.

| Scenario | Safety stock held as | Slow-mover CSL | Value (S$) | Holding cost / yr |
|---|---|---|---|---|
| A. Current rule (2 wks) | 358 fast + 242 slow FG | 59% | 90,000 | 18,000 |
| B. Pure MTS at 95% | 304 fast + 1,694 slow FG | 95% | 299,690 | 59,938 |
| C. Hybrid MTS + ATO | 304 fast FG + 311 spring units (8 types) + 444 quilt panels (15 fabric×size) | 95% | 87,569 | 17,514 |

Hybrid C serves slow movers at 95% instead of 59% for 71% less safety stock than pure MTS, at about the same cost as today (S$87,569 vs S$90,000).

Finished-goods SKUs on the shelf drop from 230 to 20, which frees most of the 45,000 sq ft and ends "stockout and excess at the same time".

Supporting changes:

- **Setup reduction** on the coil-spring and quilt-stitching machines: smaller economic batches for spring units and panels, so cycle stock shrinks too.
- **Daily assembly schedule** for ATO orders on top of the weekly MTS plan for fast movers. One planner owns the component buffers (reorder points per spring-unit type and fabric×size).
- **Raw materials**: fewer fabrics and one wire gauge mean MOQs turn quickly. Negotiate split deliveries or consignment for the remaining low-volume fabrics.

**4.3 Order fulfilment: differentiated service and information**

- The two service promises of the company are that fast moving products will be delivered ex-stock within one day, while slow moving and exclusive products will be assembled to order within 2 days. These promises match the expectations of the dealers of the company.
- Implementation of electronic ordering systems for the pharmacy. These systems allow for orders to be placed online or through EDI software instead of having to place orders via fax or phone. Also, these systems have a cutoff time for placing orders and the pharmacy has real-time visibility of the stock in their warehouse as well as the available ATP of their products. When an order is placed, the pharmacy receives a confirmation of that order.

This solution keeps the variety in the market but absorbs it at the component level. The inventory of components follows the $\sqrt{n}$ instead of $n$. The fill rate of products increases since the slow movers do not have to rely on a 59% chance of being on the shelf. The warehouse space, working capital and wastage decrease. The scheduling of the components only requires two loops each week and each day.